EL JOKER — Script du pitch, questions du jury et vidéo de démo

12 octobre 2026 · Team Chain619 · Rihem Ghoudi & Mariem Souayh

> Rihem ouvre le pitch et mène la démo ; Mariem présente la solution et la partie technique. Total : 10 minutes, 15 slides. Le texte à dire est en anglais, les consignes sont en français.

Qui parle quand

Rihem parle environ 4 min 30, Mariem environ 5 min, et vous dites la dernière slide à deux. Pendant la slide 9, Rihem lance les scénarios sur l'ordinateur pendant que Mariem explique.

#	Slide	Qui parle	Durée	Fin à	Message clé
1	Cover	Rihem	25 s	0:25	Cinq agents, une seule IA, tout prouvé sur Hedera
2	The question	Rihem	30 s	0:55	Qui contrôle une IA qui dépense de l'argent ?
3	The problem	Rihem	40 s	1:35	Hallucination, dépenses excessives, traces effaçables
4	Scenario (Gabès)	Rihem	30 s	2:05	Ici, une erreur = des médicaments qui n'arrivent pas
5	The solution	Mariem	45 s	2:50	AI proposes. Rules decide. Hedera proves.
6	The crew	Mariem	40 s	3:30	Cinq agents, un seul utilise l'IA
7	Mission flow	Mariem	40 s	4:10	De la demande à la preuve, étape par étape
8	Demo: Approved	Rihem (live)	60 s	5:10	Un vrai paiement sur testnet, 6 preuves
9	Demo: Blocked	Mariem (Rihem lance)	60 s	6:10	Quatre refus, aucun argent ne bouge
10	Human review	Rihem (live)	45 s	6:55	Au-dessus de 5 000 $, un humain décide
11	Guardian	Mariem	35 s	7:30	Pas d'IA pour décider : rien à négocier
12	Proof layer	Mariem	45 s	8:15	Seulement des empreintes sur Hedera
13	Tampering	Mariem	50 s	9:05	Une base de données peut réécrire son passé, Hedera non
14	Verified live	Rihem	40 s	9:45	Testé en vrai, feuille de route, limites honnêtes
15	Closing	Mariem puis Rihem	15 s	10:00	Autonomous, but never uncontrolled.

Si la démo live plante, restez sur les captures des slides 8 à 10 et dites : « Here is the same run, recorded live on testnet earlier today. » Ne dépassez pas 10 minutes : si vous êtes en retard à la slide 8, raccourcissez la slide 9 à deux exemples (budget et facture recyclée).

Script : slides 1 à 7

Le texte entre crochets est une consigne : ne le dites pas. Parlez lentement, environ 130 mots par minute, et regardez le jury, pas l'écran.

1. Cover — Rihem (25 s)

[Laissez l'image s'afficher, souriez, puis parlez.]

> « Good morning. We are team Chain619. I'm Rihem, and this is Mariem. This is EL JOKER. In Tunisian, *5aliha 3lina* means "leave it to us". EL JOKER is a team of five agents that runs emergency procurement for NGOs. One agent uses AI to recommend. Four are strict code that verify, decide and pay. And every decision they make is proven on Hedera. »

2. The question — Rihem (30 s)

> « AI agents are moving from talking to acting: they book, they order, they pay. The moment an agent can spend money, two questions matter. First: who is in control when it is wrong, or when someone tricks it? Second: how can anyone prove, afterwards, what it decided, and why? »

3. The problem — Rihem (40 s)

> « Today, agents fail in three ways. One: they hallucinate. They invent a supplier, a price or an approval, and act on it. Two: they overspend. They optimize the task and forget the budget, or pay the same invoice twice. Three: they hide the trail. They act first, explain later, and their logs live in a database that anyone with access can edit. So organizations have to choose: autonomy, or control. We decided to remove that trade-off. »

4. Scenario — Rihem (30 s)

> « Let's make it concrete. A water network fails in Gabès. Five hundred people need drinking water and essential medicine within 48 hours, and the NGO has five thousand dollars. Speed matters. But here, a wrong supplier, an inflated price or a double payment is not a bug. It is medicine that never arrives. Mariem will show you how EL JOKER handles it. »

5. The solution — Mariem (45 s)

> « Thank you, Rihem. Our answer fits in one line: AI proposes, rules decide, Hedera proves. We use AI where judgment helps: Claude ranks the supplier offers and explains the trade-offs. But the AI never touches the money. An independent verifier checks its work. A guardian with fixed rules decides. A human approves anything above the autonomous limit. And the payment agent pays only when all of them say yes. Every one of these steps is written to the Hedera Consensus Service. »

6. The crew — Mariem (40 s)

[Montrez chaque robot de la main en le nommant.]

> « Meet the crew, with Tunisian names. Lem3alem, "the boss", turns the request into a validated mission and never guesses a missing fact. Lemhaf, "the clever one", is the only agent powered by AI: Claude. Nasnes, "the one who watches", verifies everything independently. 3ases, "the guard", applies the

> spending rules with no AI at all. And Kanawita, "the money", pays on Hedera, and only when authorized. »

7. Mission flow — Mariem (40 s)

> « Here is one mission, end to end. The operator types the need in a chat. Lem3alem plans the mission. Lemhaf asks Claude to rank the offers. Nasnes runs ten independent checks. 3ases applies thirteen rules. If the amount is above five thousand dollars, or the risk is high, the payment waits for a human. Then Kanawita pays, once, on Hedera testnet, and every step leaves a receipt on Hedera. Rihem, show them. »

Script : slides 8 à 15

Si vous faites la démo en direct, l'application doit déjà être ouverte dans un autre onglet, avec la demande pré-remplie. Le temps que Claude réponde (environ 10 s), continuez à parler.

8. Demo: Approved — Rihem, en direct (60 s)

[Passez à l'application, appuyez sur Envoyer.]

> « This is a real run, on Hedera testnet, with Claude. I send the Gabès request. Claude compares three offers, picks Oasis at 4,150 dollars, and explains why. Nasnes passes ten out of ten checks. 3ases passes thirteen out of thirteen rules, risk low: approved. Kanawita pays 1.0375 HBAR. That is a scaled demo amount, proportional to the offer. The receipt says SUCCESS. On the right, the six decisions of this mission appear on Hedera one by one, each with a HashScan link that anyone can open. »

9. Demo: Blocked — Mariem parle, Rihem lance les scénarios (60 s)

> « A spending system is judged by what it refuses. Four examples. Budget: 100 dollars against a 4,150 dollar offer. Blocked. The money never moves, and the refusal itself is recorded on Hedera. Unverified supplier: under a 12-hour rush, the AI picks the cheapest broker. In our live run, Claude even flagged a fraud risk, but it doesn't matter, because the decision is not the AI's: Nasnes rejects, 3ases blocks. Duplicate payment: we replay exactly the same payment. Refused, still one payment. Recycled invoice: even a verified supplier cannot bill the same invoice twice. »

10. Human review — Rihem, en direct (45 s)

[Scénario E : remplissez la raison et le PIN pendant que vous parlez, puis cliquez sur Approve.]

> « Autonomy has a limit. Two thousand displaced people in Médenine need supplies. The best offer is 16,600 dollars, above our five-thousand-dollar autonomous limit. 3ases doesn't approve and doesn't block: it escalates. The payment stays on hold until an authorized operator gives a reason and a PIN, checked on the server. I approve: 4.15 HBAR paid. The approval is bound to a fingerprint of the exact facts the operator saw. If anything changes, it no longer counts. And no human can override a block. Mariem? »

11. Guardian — Mariem (35 s)

> « Why is the guardian not an AI? Because you can argue with an AI, and someone will. 3ases is plain, deterministic code: budget, verified supplier, allowed region, independent verification, invoice not reused, no previous payment, a sane price. One failed rule, and the mission is blocked. Above five thousand dollars or at high risk, a human decides. Same facts plus same policy version always give the same decision, and twenty unit tests prove it. »

12. Proof layer — Mariem (45 s)

> « How do we use Hedera? Each decision keeps its full evidence off-chain. We turn it into canonical JSON, compute a SHA-256 fingerprint, and link it to the fingerprint of the previous event: a hash chain. Only that is published on our Hedera Consensus Service topic. No names, no prices, no documents on-

> chain. Anyone can read the Mirror Node and recompute everything. And the payments are real HBAR transfers on testnet, visible on HashScan. »

13. Tampering — Mariem (50 s)

> « So why not just a database? Because a database can rewrite its past. Imagine an insider. They edit the approved price: a normal hash check catches it. But a smart insider also recomputes every hash, or simply deletes the payment record. Locally, everything looks perfect. Hedera disagrees: the consensus record says otherwise, and our verifier shows TAMPERING DETECTED. That alert is itself written to Hedera. We ran all three attacks live on testnet. Rihem? »

14. Verified live — Rihem (40 s)

> « This is not a mockup. All eight scenario checks passed live on Hedera testnet with the real Claude. Ninety-six automated tests cover the rules, the payments and the tamper detection, and every mission leaves six to eight proofs. Next: real supplier data and stablecoin settlement. Then real identity and multi-signature release. Then a field pilot with an NGO in Tunisia. Who pays? NGOs and aid programs, through a subscription plus proof volume. That is our hypothesis, not yet validated. And we are open about today's limits: fictional suppliers, scaled amounts, and a demo PIN. »

15. Closing — Mariem puis Rihem (15 s)

> **Mariem :** « AI agents should act, but never outside rules nobody can argue with, and never without proof anyone can check. »

> **Rihem :** « EL JOKER: autonomous, but never uncontrolled. The code and the Hedera topic are public, so you can verify every decision yourself. Thank you. Shoukran. »

[Restez sur la dernière slide pendant les questions : les liens GitHub et HashScan y sont affichés.]

Questions du jury

Répondez en 20 secondes maximum : une phrase de réponse, une preuve, et c'est tout. Si vous ne savez pas, dites « Good question, we haven't validated that yet » plutôt que d'inventer. Règle de partage : Mariem répond à l'IA, au Guardian et à Hedera ; Rihem répond à la démo, au marché et à la suite du projet.

Hedera

Why Hedera and not a normal database? (Mariem)
« A database can be rewritten by whoever controls it, including us. Hedera gives every decision an ordered, timestamped record that we cannot change, and anyone can check it on the Mirror Node. Slide 13 shows the attack that only Hedera catches. »

Why Hedera and not another blockchain? (Mariem)
« We need many small proofs per mission: six to eight messages each. Hedera's Consensus Service gives fixed, very low fees and finality in a few seconds, so proving every decision stays cheap and fast. »
[Avant le pitch, vérifiez le prix exact d'un message HCS sur hedera.com/fees pour pouvoir citer un chiffre.]

Why not put all the data on-chain? (Mariem)
« Privacy. Beneficiary data, prices and invoices stay off-chain. On Hedera we only publish fingerprints, under 1 KB per message. That is enough to prove nothing changed, without exposing anyone. »

What if Hedera is down or slow? (Mariem)
« Then nobody gets paid. Kanawita waits until the Guardian's approval is anchored on Hedera before paying. We retry the anchoring, and if a receipt is lost we query the same transaction again instead of sending a new one. »

Is this real or simulated? (Rihem)
« Real, on testnet. The payment you saw has a transaction ID you can open on HashScan, and our topic 0.0.11007004 is public. All eight scenario checks passed live with the real Claude. »

L'IA et la sécurité

What if the AI hallucinates or gets prompt-injected? (Mariem)
« Then it recommends something wrong, and that's all it can do. The AI never decides and never pays. Nasnes checks its claims against the registry, and 3ases blocks anything that breaks a rule. We even hide the registry status from the AI, so it can't just copy the answer. »

Can the AI change the amount paid? (Mariem)
« No. The amount comes from the verified offer, and the payment amount is computed by code, never by the model. »

What happens if Claude is unavailable? (Mariem)
« After 45 seconds we switch to a simple rule-based ranking, clearly labeled FALLBACK in the interface and in the proofs. The rules and the payment checks don't change. »

Why five agents instead of one? (Mariem)
« Separation of duties, like in a real organization: the person who chooses the supplier is not the one who approves or pays. Each agent has one job, and only one of them uses AI. »

Who writes the 13 rules, and can someone change them quietly? (Mariem)
« The rules live in a versioned policy file. Its version includes a fingerprint of the file, and that version is written into every proof. If someone changes a rule, it shows on Hedera. »

How do you prevent paying twice? (Rihem)
« Three layers: an idempotency key per payment, database constraints that allow one live payment per mission and per invoice, and a check that the invoice number was never used before. In the demo, replaying the payment returns an HTTP 409. »

Can an insider fake a human approval? (Rihem)
« The approval needs an authorized operator and a PIN checked on the server; five wrong PINs lock the mission for a minute. The approval is tied to a fingerprint of the exact facts reviewed, and it is recorded on Hedera. In production we would add real identity and multi-signature. »

Marché et suite

Who are your customers, and who pays? (Rihem)
« NGOs, humanitarian organizations and grant programs that buy in emergencies. The model we imagine is a subscription per organization plus a fee per proof. That is a hypothesis: we haven't validated it with customers yet. »

Have you talked to a real NGO? (Rihem)
« Not yet. The suppliers in the demo are fictional. Our next step is a field pilot with an NGO in Tunisia. »
[Si vous avez parlé à quelqu'un d'ici le pitch, remplacez cette réponse par ce qui s'est vraiment passé.]

Why HBAR and not a stablecoin? (Rihem)
« HBAR was the simplest way to show a real, confirmed payment on testnet. The amounts are scaled for the demo. Stablecoin settlement is the first item on our roadmap. »

Who are your competitors? (Rihem)
« Agent frameworks and payment APIs give autonomy, and audit tools give logs. We combine both: hard spending rules that no AI can talk around, and a public proof anyone can check. »

Does it scale? (Mariem)
« Each mission sends six to eight messages to Hedera, which is designed for very high throughput. The demo uses SQLite; for production we would move to a server database. The agent logic doesn't change. »

What would you do with three more months? (Rihem)
« Real supplier data, stablecoin settlement, real identity with multi-signature release, and a pilot with one NGO. »

What was the hardest part? (Rihem ou Mariem)
« A live testnet bug: a receipt timed out after the message was already accepted, so our retry created a duplicate proof and triggered a false tampering alarm. We fixed it by reading the receipt of the same transaction instead of resending, and added a test for it. »

Vidéo de démo (3 minutes)

La vidéo montre cinq moments, dans cet ordre : un paiement approuvé, une tentative de double paiement, un fournisseur piège, une validation humaine et une falsification détectée. Rihem fait la voix des parties démo et Mariem celle des pièges et de Hedera. Vous pouvez enregistrer les voix séparément et les poser au montage.

Temps	À l'écran	Action	Voix off
0:00–0:12	Couverture du pitch	Aucune	**Rihem :** « Meet EL JOKER: five agents that buy emergency supplies for NGOs. One AI recommends. Four rule-based agents verify, decide and pay. Every decision is proven on Hedera. »
0:12–0:20	Page d'accueil de l'application	Clic sur « Go for EL JOKER »	**Rihem :** « This is a real run, on Hedera testnet, with Claude. »
0:20–0:55	Conversation, scénario A pré-rempli, badge « AI · claude-opus-5-5 » en haut	Zoom sur le badge, clic sur Envoyer ; couper l'attente au montage	**Rihem :** « An NGO needs water and medicine for 500 people in Gabès, with a 5,000 dollar budget. Claude compares three offers and picks Oasis at 4,150 dollars. Nasnes checks ten facts. 3ases applies thirteen rules: approved. »
0:55–1:15	Message de paiement et panneau des preuves Hedera	Clic sur « View on HashScan », montrer SUCCESS sur HashScan	**Rihem :** « Kanawita pays 1.0375 HBAR on testnet. The receipt is confirmed, and anyone can check it on HashScan. Six decisions are anchored on the Hedera Consensus Service. »
1:15–1:30	Panneau Actions de la même mission	Clic sur « Replay payment (attack) », montrer le refus	**Mariem :** « Someone sends the exact same payment again. Rejected. There is still exactly one payment. »
1:30–1:55	Puce « Scenario D: Unverified supplier »	Clic sur la puce, Envoyer, montrer BLOCKED	**Mariem :** « Now a trap. Under time pressure, the AI picks a cheap broker that isn't in the registry. It doesn't matter: Nasnes rejects it, 3ases blocks it, and no money moves. The refusal is proven on Hedera too. »
1:55–2:25	Scénario E, carte d'approbation humaine	Choisir l'opérateur, écrire une raison, entrer le PIN, clic sur « Approve »	**Rihem :** « Above 5,000 dollars, a human decides. The payment waits for an authorized operator, a reason and a PIN, checked on the server. Approved: 4.15 HBAR paid. »
2:25–2:50	Onglet « Proof & verify »	« Verify against Hedera » → VALID ; « Insider rewrites hashes too » ; « Verify against Hedera » → TAMPERING DETECTED	**Mariem :** « Finally, an insider edits the database and recomputes every hash. Locally, it looks perfect. But Hedera remembers: the verifier says TAMPERING DETECTED. »
2:50–3:00	Slide finale du pitch	Aucune	**Les deux :** « EL JOKER: autonomous, but never uncontrolled. Code on GitHub, proofs on Hedera. »

Les boutons de falsification restent grisés tant que toutes les preuves ne sont pas ancrées : attendez quelques secondes après le paiement du scénario E avant de cliquer.

Avant d'enregistrer

☐ Lancer la démo en mode live et vérifier que le badge en haut affiche « AI · claude-opus-5-5 », pas « AI · fallback ».

☐ Vérifier qu'il reste des HBAR testnet sur le compte opérateur et des crédits sur le compte Anthropic.
☐ Faire une répétition complète sans enregistrer, pour connaître les temps d'attente.
☐ Navigateur en plein écran (1920×1080, zoom 110 %), mode Ne pas déranger de Windows activé, onglets inutiles fermés.
☐ Ne jamais montrer le fichier .env, un terminal avec des clés, ni le portail Hedera.
☐ Enregistrer avec la Xbox Game Bar (Win + Alt + R) ou OBS, micro de casque, pièce calme.
☐ Au montage : couper les attentes, zoomer sur les badges et sur TAMPERING DETECTED, ajouter des sous-titres en anglais.
☐ Exporter en MP4 1080p et vérifier la durée maximale imposée par le hackathon.